Project Proposals for MSC in AI (Conversion)

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I have two project proposals for MSc in AI (Conversion)

Project 1

Project title

Deception Detection using Computational Linguistic Analysis for statement and conversation analysis.

Project outline:
Detecting the deceptive cues in the conversations and statements is one of the challenging tasks. Experts are usually trained for interrogation methods focusing on linguistic analysis. In literature, there are a few computational models, including AI systems have been built to identify the deceptive cues in the conversations and written statements. In addition to verify of applications, these techniques largely support law enforcement departments and credibility assessments in businesses. However, identifying deceptive cues using AI systems with ever changing environment and platforms for communication is still a difficult problem.

In this project, student will start with available open-source datasets (Mafiascum [1]) with available computation methods with comparative analysis and finally design and implement one of the deeplearning-based model.

[1] The Mafiascum Dataset: A Large Text Corpus for Deception Detection
https://arxiv.org/abs/1811.07851

Impact and outcomes:
Other than a publication, the results of this project could help the law enforcement departments and research community.

Required skill sets:
Essential: Programming (Python), Computation analysis (Statistics), Machine learning and Deeplearning – training and testing models.
Desired but not essential: Computational Linguistic Analysis

Skills that the student will develop during the project:
The project will equip students with skills to formulate problems, conduct experiments, systematically analyse data, and report results effectively.

Project 2

Project title

Validation of Saliency Maps of Deeplearning models for Biomedical Signals such as ECG and EEG.

Project outline:
In the intersection of AI and Biomedical, researchers often train large deeplearning models to predict a category of the signal source or phenotypes. After achieving a well performing model, researchers use saliency maps as one of the methods, to produce explanability of the models. Saliency maps highlight the segments of input signals showing their importance for the predictability. However, there is a lack of systematic validation of these saliency maps, especially for, biomedical signals such as Electrocardiogram (ECG). This project explores and validates the saliency maps generated by trained deeplearning models. Specifically, we aim to identify which saliency map method works well, if they even work and where they fail.

Impact and outcomes:
Other than a publication, the results of this project could help the research community greatly by providing the guidelines for using and interpreting saliency maps correctly.

Required skill sets:
Essential: Programming (Python), Computation analysis (Statistics), Machine learning and Deeplearning – training and testing models.
Desired but not essential: Signal Processing, Information Theory, Mathematical Modelling

Skills that the student will develop during the project:
The project will equip students with skills to formulate problems, conduct experiments, systematically analyse data, and report results effectively.

My research interests include biomedical systems design and analysis (electrophysiology), deception detection, signal processing, mathematical modelling, information theory. More details on my past projects and publications can be found at: https://nikeshbajaj.in/research/.

If you are interested in one of the above projects or you your own project idea, feel free to contact at nikesh.bajaj@qmul.ac.uk